

BEAM SUPPORTED BY ROD HANGERS

MAX CANT. MOMENT - M

$$-M_{DL} - P_1, 15^{\#} \times 3.1 = 47^{\#}$$

$$P_{38B}, 30 \times (2.25 + 1.0) = 98$$

$$w; 16 \times 3.1^2 / 2 = 77$$

$$-M_{DL} = 222^{\#}$$

$$-M_{LL} - w, 60 \times 3.1^2 / 2 = 288^{\#}$$

$$-M_{TL} = 510$$

1.5 x 3.1

USE ALTERNATE LL - $P_{LL} = 300^{\#}$

$$-M_{LL} = 300 \times 3.1 = 930^{\#}$$

$$-M_{DL} = 222^{\#}$$

$$-M_{TL} = 1152^{\#} \leftarrow \text{GOVERNS}$$

CHECK 2 - 2 x 6 BEAM FOR $-M_{TL}$ MAX.

$$f_b = 12 \times 1152 / (2 \times 7.56) = 914^{\#}/0^{\#} \underline{\text{OK}}$$

$$f_w = 1.5 \times (300 + 15 + 30 + 30 + 3.1 \times 16) / 8.25 = 77^{\#}/0^{\#} \underline{\text{OK}}$$

MAX + MOMENT - DL ON CANT - $P_{LL} = 300^{\#}$ AT CTR SPAN
+ M_{DL} : (W/O CANT)

$$+ 138 \times 3 = 414^{\#}$$

$$- w_{DL}, 16 \times 3^2 / 2 = -72$$

$$- P_{38B}, 30 \times 2.75 = -83$$

$$- P_{60B}, 60 \times 1.0 = -60^{\#}$$

$$+ M_{DL} = 199^{\#} \text{ (W/O CANT)}$$

$$+ M_{LL} - P = 300^{\#} \times 6/4 = 450^{\#} \text{ " "}$$

$$+ M_{TL} = 649^{\#} \text{ (W/O CANT)}$$